A central limit theorem for the determinant of a Wigner matrix

Terence Tao · 28 November 2011

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Van Vu and I have just uploaded to the arXiv our paper A central limit theorem for the determinant of a Wigner matrix, submitted to Adv. Math.. It studies the asymptotic distribution of the determinant $\det M_n$ of a random Wigner matrix (such as a matrix drawn from the Gaussian Unitary Ensemble (GUE) or Gaussian Orthogonal Ensemble (GOE)).

Before we get to these results, let us first discuss the simpler problem of studying the determinant $\det A_n$ of a random *iid* matrix $A_n = (\zeta_{ij})_{1 \leq i,j \leq n}$, such as a real gaussian matrix (where all entries are independently and identically distributed using the standard real normal distribution $\zeta_{ij} \equiv N(0,1)_{\mathbf{R}}$), a complex gaussian matrix (where all entries are independently and identically distributed using the standard complex normal distribution $\zeta_{ij} \equiv N(0,1)_{\mathbf{C}}$, thus the real and imaginary parts are independent with law $N(0,1/2)_{\mathbf{R}}$), or the random sign matrix (in which all entries are independently and identically distributed according to the Bernoulli distribution $\zeta_{ij} \equiv \pm 1$ (with a $1/2$ chance of either sign). More generally, one can consider a matrix A_n in which all the entries ζ_{ij} are independently and identically distributed with mean zero and variance 1.

We can expand $\det A_n$ using the Leibniz expansion

$$\det A_n = \sum_{\sigma \in S_n} I_\sigma, \tag{1}$$

where $\sigma : \{1,...,n\} \to \{1,...,n\}$ ranges over the permutations of $\{1,...,n\}$, and I_σ is the product

$$I_\sigma := \mathrm{sgn}(\sigma) \prod_{i=1}^n \zeta_{i\sigma(i)}.$$

From the iid nature of the ζ_{ij}, we easily see that each I_σ has mean zero and variance one, and are pairwise uncorrelated as σ varies. We conclude that $\det A_n$ has mean zero and variance $n!$ (an observation first made by Turán). In particular, from Chebyshev's inequality we see that $\det A_n$ is typically of size $O(\sqrt{n!})$.

It turns out, though, that this is not quite best possible. This is easiest to explain in the real gaussian case, by performing a computation first made by Goodman. In this case, $\det A_n$ is clearly symmetrical, so we can focus attention on the magnitude $|\det A_n|$. We can interpret this quantity geometrically as the volume of an n-dimensional parallelopiped whose generating vectors $X_1,...,X_n$ are independent real gaussian vectors in $\mathbf{R}^n$ (i.e. their coefficients are iid with law $N(0,1)_{\mathbf{R}}$). Using the classical base-times-height formula, we thus have

$$|\det A_n| = \prod_{i=1}^n \mathrm{dist}(X_i, V_i) \tag{2}$$

where V_i is the $i-1$-dimensional linear subspace of $\mathbf{R}^n$ spanned by $X_1,...,X_{i-1}$ (note that $X_1,...,X_n$, having an absolutely continuous joint distribution, are almost surely linearly independent). Taking logarithms, we conclude

$$\log|\det A_n| = \sum_{i=1}^n \log \mathrm{dist}(X_i, V_i).$$

Now, we take advantage of a fundamental symmetry property of the Gaussian vector distribution, namely its invariance with respect to the orthogonal group $O(n)$. Because of this, we see that if we fix

$X_1, \ldots, X_{i-1}$ (and thus V_i, the random variable $\mathrm{dist}(X_i, V_i)$ has the same distribution as $\mathrm{dist}(X_i, \mathbf{R}^{i-1})$, or equivalently the χ distribution

$$\chi_{n-i+1} := (\sum_{j=1}^{n-i+1} \xi_{n-i+1,j}^2)^{1/2}$$

where $\xi_{n-i+1,1}, \ldots, \xi_{n-i+1,n-i+1}$ are iid copies of $N(0,1)_{\mathbf{R}}$. As this distribution does not depend on the $X_1, \ldots, X_{i-1}$, we conclude that the law of $\log|\det A_n|$ is given by the sum of n independent χ-variables:

$$\log|\det A_n| \equiv \sum_{j=1}^{n} \log \chi_j.$$

A standard computation shows that each χ_j^2 has mean j and variance $2j$, and then a Taylor series (or Ito calculus) computation (using concentration of measure tools to control tails) shows that $\log \chi_j$ has mean $\frac{1}{2}\log j - \frac{1}{2j} + O(1/j^{3/2})$ and variance $\frac{1}{2j} + O(1/j^{3/2})$. As such, $\log|\det A_n|$ has mean $\frac{1}{2}\log n! - \frac{1}{2}\log n + O(1)$ and variance $\frac{1}{2}\log n + O(1)$. Applying a suitable version of the central limit theorem, one obtains the asymptotic law

$$\frac{\log|\det A_n| - \frac{1}{2}\log n! + \frac{1}{2}\log n}{\sqrt{\frac{1}{2}\log n}} \to N(0,1)_{\mathbf{R}}, \tag{3}$$

where $\to$ denotes convergence in distribution. A bit more informally, we have

$$|\det A_n| \approx n^{-1/2}\sqrt{n!}\exp(N(0, \log n/2)_{\mathbf{R}}) \tag{4}$$

when A_n is a real gaussian matrix; thus, for instance, the median value of $|\det A_n|$ is $n^{-1/2+o(1)}\sqrt{n!}$. At first glance, this appears to conflict with the second moment bound $\mathbf{E}|\det A_n|^2 = n!$ of Turán mentioned earlier, but once one recalls that $\exp(N(0,t)_{\mathbf{R}})$ has a second moment of $\exp(2t)$, we see that the two facts are in fact perfectly consistent; the upper tail of the normal distribution in the exponent in (4) ends up dominating the second moment.

It turns out that the central limit theorem (3) is valid for any real iid matrix with mean zero, variance one, and an exponential decay condition on the entries; this was first claimed by Girko, though the arguments in that paper appear to be incomplete. Another proof of this result, with more quantitative bounds on the convergence rate has been recently obtained by Hoi Nguyen and Van Vu. The basic idea in these arguments is to express the sum in (2) in terms of a martingale and apply the martingale central limit theorem.

If one works with complex gaussian random matrices instead of real gaussian random matrices, the above computations change slightly (one has to replace the real χ distribution with the complex χ distribution, in which the $\xi_{i,j}$ are distributed according to the complex gaussian $N(0,1)_{\mathbf{C}}$ instead of the real one). At the end of the day, one ends up with the law

$$\frac{\log|\det A_n| - \frac{1}{2}\log n! + \frac{1}{4}\log n}{\sqrt{\frac{1}{4}\log n}} \to N(0,1)_{\mathbf{R}}, \tag{5}$$

or more informally

$$|\det A_n| \approx n^{-1/4}\sqrt{n!}\exp(N(0, \log n/4)_{\mathbf{R}}) \tag{6}$$

(but note that this new asymptotic is still consistent with Turán's second moment calculation).

We can now turn to the results of our paper. Here, we replace the iid matrices A_n by *Wigner matrices* $M_n = (\zeta_{ij})_{1 \leq i,j \leq n}$, which are defined similarly but are constrained to be Hermitian (or real symmetric), thus $\zeta_{ij} = \overline{\zeta_{ji}}$ for all i, j. Model examples here include the Gaussian Unitary Ensemble (GUE), in which $\zeta_{ij} \equiv N(0,1)_{\mathbf{C}}$ for $1 \leq i < j \leq n$ and $\zeta_{ij} \equiv N(0,1)_{\mathbf{R}}$ for $1 \leq i = j \leq n$, the Gaussian Orthogonal Ensemble (GOE), in which $\zeta_{ij} \equiv N(0,1)_{\mathbf{R}}$ for $1 \leq i < j \leq n$ and $\zeta_{ij} \equiv N(0,2)_{\mathbf{R}}$ for $1 \leq i = j \leq n$, and the *symmetric Bernoulli ensemble*, in which $\zeta_{ij} \equiv \pm 1$ for $1 \leq i \leq j \leq n$ (with probability

$1/2$ of either sign). In all cases, the upper triangular entries of the matrix are assumed to be jointly independent. For a more precise definition of the Wigner matrix ensembles we are considering, see the introduction to our paper.

The determinants $\det M_n$ of these matrices still have a Leibniz expansion. However, in the Wigner case, the mean and variance of the I_σ are slightly different, and what is worse, they are not all pairwise uncorrelated any more. For instance, the mean of I_σ is still usually zero, but equals $(-1)^{n/2}$ in the exceptional case when σ is a perfect matching (i.e. the union of exactly $n/2$ 2-cycles, a possibility that can of course only happen when n is even). As such, the mean $\mathbf{E} \det M_n$ still vanishes when n is odd, but for even n it is equal to

$$(-1)^{n/2} \frac{n!}{(n/2)! 2^{n/2}}$$

(the fraction here simply being the number of perfect matchings on n vertices). Using Stirling's formula, one then computes that $|\mathbf{E} \det A_n|$ is comparable to $n^{-1/4}\sqrt{n!}$ when n is large and even. The second moment calculation is more complicated (and uses facts about the distribution of cycles in random permutations, mentioned in this previous post), but one can compute that $\mathbf{E}|\det A_n|^2$ is comparable to $n^{1/2} n!$ for GUE and $n^{3/2} n!$ for GOE. (The discrepancy here comes from the fact that in the GOE case, I_σ and I_ρ can correlate when ρ contains reversals of k-cycles of σ for $k \geq 3$, but this does not happen in the GUE case.) For GUE, much more precise asymptotics for the moments of the determinant are known, starting from the work of Brezin and Hikami, though we do not need these more sophisticated computations here.

Our main results are then as follows.

> **Theorem 1** Let M_n be a Wigner matrix.
>
> - If M_n is drawn from GUE, then
>
> $$\frac{\log|\det M_n| - \frac{1}{2}\log n! + \frac{1}{4}\log n}{\sqrt{\frac{1}{2}\log n}} \to N(0,1)_{\mathbf{R}}.$$
>
> - If M_n is drawn from GOE, then
>
> $$\frac{\log|\det M_n| - \frac{1}{2}\log n! + \frac{1}{4}\log n}{\sqrt{\log n}} \to N(0,1)_{\mathbf{R}}.$$
>
> - The previous two results also hold for more general Wigner matrices, assuming that the real and imaginary parts are independent, a finite moment condition is satisfied, and the entries match moments with those of GOE or GUE to fourth order. (See the paper for a more precise formulation of the result.)

Thus, we informally have

$$|\det M_n| \approx n^{-1/4}\sqrt{n!}\exp(N(0, \log n/2)_{\mathbf{R}})$$

when M_n is drawn from GUE, or from another Wigner ensemble matching GUE to fourth order (and obeying some additional minor technical hypotheses); and

$$|\det M_n| \approx n^{-1/4}\sqrt{n!}\exp(N(0, \log n)_{\mathbf{R}})$$

when M_n is drawn from GOE, or from another Wigner ensemble matching GOE to fourth order. Again, these asymptotic limiting distributions are consistent with the asymptotic behaviour for the second moments.

The extension from the GUE or GOE case to more general Wigner ensembles is a fairly routine application of the *four moment theorem* for Wigner matrices, although for various technical reasons we do not quite use the existing four moment theorems in the literature, but adapt them to the log determinant. The main idea is to express the log-determinant as an integral

$$\log|\det M_n| = \frac{1}{2} n \log n - n \ \mathrm{Im} \int_0^\infty s(\sqrt{-1}\eta)\ d\eta \tag{7}$$

of the Stieltjes transform

$$s(z) := \frac{1}{n}\ \mathrm{tr}(\frac{1}{\sqrt{n}} M_n - z)^{-1}$$

of M_n. Strictly speaking, the integral in (7) is divergent at infinity (and also can be ill-behaved near zero), but this can be addressed by standard truncation and renormalisation arguments (combined with known facts about the least singular value of Wigner matrices), which we omit here. We then use a variant of the four moment theorem for the Stieltjes transform, as used by Erdos, Yau, and Yin (based on a previous four moment theorem for individual eigenvalues introduced by Van Vu and myself). The four moment theorem is proven by the now-standard Lindeberg exchange method, combined with the usual resolvent identities to control the behaviour of the resolvent (and hence the Stieltjes transform) with respect to modifying one or two entries, together with the delocalisation of eigenvector property (which in turn arises from local semicircle laws) to control the error terms.

Somewhat surprisingly (to us, at least), it turned out that it was the first part of the theorem (namely, the verification of the limiting law for the invariant ensembles GUE and GOE) that was more difficult than the extension to the Wigner case. Even in an ensemble as highly symmetric as GUE, the rows are no longer independent, and the formula (2) is basically useless for getting any non-trivial control on the log determinant. There is an explicit formula for the joint distribution of the eigenvalues of GUE (or GOE), which does eventually give the distribution of the cumulants of the log determinant, which then gives the required central limit theorem; but this is a lengthy computation, first performed by Delannay and Le Caer.

Following a suggestion of my colleague, Rowan Killip, we give an alternate proof of this central limit theorem in the GUE and GOE cases, by using a beautiful observation of Trotter, namely that the GUE or GOE ensemble can be conjugated into a tractable tridiagonal form. Let me state it just for GUE:

Proposition 2 (Tridiagonal form of GUE) Let M'_n be the random tridiagonal real symmetric matrix

$$M'_n = \begin{pmatrix} a_1 & b_1 & 0 & \ldots & 0 & 0 \\ b_1 & a_2 & b_2 & \ldots & 0 & 0 \\ 0 & b_2 & a_3 & \ldots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \ldots & a_{n-1} & b_{n-1} \\ 0 & 0 & 0 & \ldots & b_{n-1} & a_n \end{pmatrix}$$

where the $a_1, \ldots, a_n, b_1, \ldots, b_{n-1}$ are jointly independent real random variables, with $a_1, \ldots, a_n \equiv N(0,1)_{\mathbf{R}}$ being standard real Gaussians, and each b_i having a χ-distribution:

$$b_i = (\sum_{j=1}^{i} |z_{i,j}|^2)^{1/2}$$

where $z_{i,j} \equiv N(0,1)_{\mathbf{C}}$ are iid complex gaussians. Let M_n be drawn from GUE. Then the joint eigenvalue distribution of M_n is identical to the joint eigenvalue distribution of M'_n.

Proof: Let M_n be drawn from GUE. We can write

$$M_n = \begin{pmatrix} M_{n-1} & X_n \\ X_n^* & a_n \end{pmatrix}$$

where M_{n-1} is drawn from the $n-1 \times n-1$ GUE, $a_n \equiv N(0,1)_{\mathbf{R}}$, and $X_n \in \mathbf{C}^{n-1}$ is a random gaussian vector with all entries iid with distribution $N(0,1)_{\mathbf{C}}$. Furthermore, M_{n-1}, X_n, a_n are jointly independent.

We now apply the tridiagonal matrix algorithm. Let $b_{n-1} := |X_n|$, then b_n has the χ-distribution indicated in the proposition. We then conjugate M_n by a unitary matrix U that preserves the final basis vector e_n, and maps X to $b_{n-1}e_{n-1}$. Then we have

$$UM_nU^* = \begin{pmatrix} \widetilde{M}_{n-1} & b_{n-1}e_{n-1} \\ b_{n-1}e_{n-1}^* & a_n \end{pmatrix}$$

where $\widetilde{M}_{n-1}$ is conjugate to M_{n-1}. Now we make the crucial observation: because M_{n-1} is distributed according to GUE (which is a unitarily invariant ensemble), and U is a unitary matrix independent of M_{n-1}, $\widetilde{M}_{n-1}$ is also distributed according to GUE, and remains independent of both b_{n-1} and a_n.

We continue this process, expanding UM_nU^* as

$$\begin{pmatrix} M_{n-2} & X_{n-1} & 0 \\ X_{n-1}^* & a_{n-1} & b_{n-1} \\ 0 & b_{n-1} & a_n. \end{pmatrix}$$

Applying a further unitary conjugation that fixes e_{n-1}, e_n but maps X_{n-1} to $b_{n-2}e_{n-2}$, we may replace X_{n-1} by $b_{n-2}e_{n-2}$ while transforming M_{n-2} to another GUE matrix $\widetilde{M}_{n-2}$ independent of $a_n, b_{n-1}, a_{n-1}, b_{n-2}$. Iterating this process, we eventually obtain a coupling of M_n to M'_n by unitary conjugations, and the claim follows. □

The determinant of a tridiagonal matrix is not quite as simple as the determinant of a triangular matrix (in which it is simply the product of the diagonal entries), but it is pretty close: the determinant D_n of the above matrix is given by solving the recursion

$$D_i = a_i D_{i-1} + b_{i-1}^2 D_{i-2}$$

with $D_0 = 1$ and $D_{-1} = 0$. Thus, instead of the product of a sequence of independent scalar χ distributions as in the gaussian matrix case, the determinant of GUE ends up being controlled by the product of a sequence of independent 2×2 matrices whose entries are given by gaussians and χ distributions. In this case, one cannot immediately take logarithms and hope to get something for which the martingale central limit theorem can be applied, but some *ad hoc* manipulation of these 2×2 matrix products eventually does make this strategy work. (Roughly speaking, one has to work with the logarithm of the Frobenius norm of the matrix first.)

Comments (13)

Greg Martin · 28 November 2011

Tiny typo, near the beginning when Chebyshev's inequality is invoked: I think it should be $O(\sqrt{n!})$, with a factorial. *[Corrected, thanks – T.]*

Florent Benaych-Georges · 29 November 2011

Tiny typo : "Brezis" should be "Brezin".

Peter Eichelsbacher · 29 November 2011

Tiny typos: in Theorem 3, Wigner matrix obeying Condition C1 (not C0),
and in the statement for the second moment: If M_n is drawn (twice). *[Thanks, this will be corrected in the next revision of the ms – T.]*

Xavier Gabaix · 29 November 2011

Tiny typo: right below (7), the definition of the Stieltjes transform has M in the numerator. *[Corrected, thanks – T.]*

yunqing2010 · 29 November 2011

Does the right hand of (7) miss a term like

$$\log\left|det(\frac{M_n}{\sqrt{n}}-\sqrt{-1}\infty)\right|$$

?

> **Terence Tao** · 29 November 2011
>
> Yes, one needs to truncate and renormalise the integral in order to eliminate the $\log\infty$ type divergence. This is discussed in detail in the paper, but I am omitting it in this post as it is a minor technical issue rather than a principal difficulty in the argument.

Jayadev Athreya · 29 November 2011

In Theorem, 1, should it be M_n instead of A_n? *[Corrected, thanks – T.]*

Craig · 1 December 2011

Tiny typo: Below (4), shouldn't it be the median value of

$$|detA_n|$$

, not

$$\log|detA_n|$$

? *[Corrected, thanks – T.]*

Anonymous · 1 December 2011

Tiny typos: "gaussian" –> "Gaussian" (A total of 12 time if my counting is correct; 13 including the tags.)

Antonio · 2 December 2011

i Professor Tarece Tao. I am a student of thirteen years and I love math and, therefore, gives great
I would do the math. I wanted to ask whether Godel's incompleteness theorems imply that in
mathematics there are many problems unsolved? if it is true, then there are infinitely many complex

problems as the 'Riemann Hypothesis, or even more complex? Thanks Professor, thanks for all she is my idol, I wanted to also ask where can I buy your books? Sorry for my poor English because I'm Italian.



kmkosinski · 5 December 2011

The conclusion from (2) is that $|detA_n| = \prod_{j=1}^{n} \chi_j$, where χ_j are independent χ-variables. That is, $\chi_i^2 = \sum_{j=1}^{i} X_{i,j}^2$ and $\{X_{i,j}\}$ are iid such that $X_{i,j} = Z_{i,j}^2$, $\{Z_{i,j}\}$ $N(0,1)$ distributed.
Hence, $|\det A_n|^2 = \prod_{k=1}^{n} S_k$, where $S_k = X_{k,1} + ... + X_{k,k}$. That is, $|\det A_n|^2$ can be seen as a product of independent sums of iid random variables. Central limit theorem for such products (and its connection to Wishart determinants) has been discussed in G. Rempala, J. Wesolowski "Asymptotics for products of independent sums with an application to Wishart determinants", Stat. Probab. Lett. 74 (2005) 129–138.
In particular, in this paper, it has been shown that for any sequence of iid positive square integrable random variables $\{X_{i,j}\}$ with finite absolute moment of order $p > 2$
$\left(\frac{n^{\gamma^2/2}}{n!\mu^n} \prod_{k=1}^{n} S_k\right)^{1/(\gamma\sqrt{\log n})}$

tends in distribution to $\exp(N(0,1))$, where $\mu = EX_{1,1}$, $\sigma^2 = Var(X_{1,1})$ and $\gamma = \sigma/\mu$ is the coefficient of variation. In this case we have $X_{i,j} = Z_{i,j}^2$ so that $\mu = 1$, $\sigma^2 = 2$, $\gamma = \sqrt{2}$. From this, clearly the result in (3) follows. The assumption on $p > 2$ was later removed in K. Kosinski "Asymptotics for sums of a function of normalized independent sums, Stat. Probab. Lett. 79 (2009) 415–419, in the sense that a different normalization than $n^{\gamma^2/2}$ is needed in the case $p = 2$.
Various other generalizations have been considered for asymptotics for products of (not necessarily independent) sums of random variables.



Anonymous · 9 December 2011

I am new to math research: what's the significance of this? Should the references be included in the paper by professor Tao? Is the paper by prof. Tao completely derivative or not (is there an original contribution)?



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[...] paper), as well as central limit theorems for log-determinants of Wigner matrices (as we did in this other recent paper). In both of these papers, the main idea was to use the Four Moment Theorem for Wigner matrices [...]